Untitled2 | Untitled1 | codebasics | EDAProject

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Code

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[6]:
```
#**1. Write a program to find the length of the string without using inbuilt function (len)**

l=[1,2,3,4,5,6,0,5]
count =0
for i in l:
    count += 1
print(count)
```

8

[10]:
```
#**2. Write a program to reverse a string without using any inbuilt functions.**
s='Vaibhav'
rev=""
for i in range(len(s)-1,-1,-1):
    rev += s[i]
print(rev)
```

vahbiaV

[8]:
```
s='Vaibhav'
s[::-1]
```

[8]: 'vahbiaV'

[12]:
```
#**3. Write a program to replace one string with another. e.g. "Hello World" replaces "World" with "Universe".**

s= "Hello World"
s1=s.replace("World","Universe")
```

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vahbiaV

```
[8]: s='Vaibhav'
     s[::-1]

[8]: 'vahbiaV'

[12]: #**3. Write a program to replace one string with another. e.g. "Hello World" replaces "World" with "Universe".**

      s= "Hello World"
      s1=s.replace("World","Universe")
      print(s1)

      Hello Universe

[18]: #**4. How to convert a string to a list and vice-versa.**

      #string to a list
      v='Vaibhav'
      v1=v.split()
      print(v1)

      text ="hello world"
      list_of_chars=list(text)
      print(list_of_chars)

      #list to string
      chars=['h','e','l','l','o']
      new_chars="".join(chars)
      print(new_chars)

      ['Vaibhav']
      ['h', 'e', 'l', 'l', 'o', ' ', 'w', 'o', 'r', 'l', 'd']
      hello

[ ]: #**5. Convert the string "Hello welcome to Python" to a comma separated string.**

      b="Hello welcome to Python"
```

```
hello
```

[26]:

```
#**5. Convert the string "Hello welcome to Python" to a comma separated string.**

b="Hello welcome to Python"
b1=b.split()
b2=",".join(b1)
print(b2)

c=""
for i in b:
    if i == " ":
        c+=","
    else:
        c+=i
print(c)
```

```
Hello,welcome,to,Python
Hello,welcome,to,Python
```

[28]:

```
#**6. Write a program to print alternate characters in a string.**

p="python"
print(p[::2])
print(p[1::2])
```

```
pto
yhn
```

[29]:

```
#**7. Write a Program to print ascii values of the characters present in a string.**
s = input("Enter a string: ")

for char in s:
    print(char, "→", ord(char))
```

```
Enter a string:  vaibhav
v → 118
a → 97
i → 105
```

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Code

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```
a → 97
v → 118
```

[30]:
```
#**8. Write a function to convert upper case to lower case and vice-versa without using inbuilt methods.**

s="vaibhav VAIBHAV"
res=""
for s1 in s:
    if 'a' <= s1 <= 'z':
        res+=chr(ord(s1)-32)
    elif 'A' <= s1 <= 'Z':
        res+=chr(ord(s1)+32)
    else:
        res+=s1

print(res)
```

```
VAIBHAV vaibhav
```

[31]:
```
#**9. Write a program to swap two numbers without using the 3rd variable.**
a = 10
b = 20

a = a + b
b = a - b
a = a - b

print(a, b)
```

```
20 10
```

```
#**10. Write a program to merge two different lists.**
list1=[1,2,3]
list2=[3,4,5]

merged_list=list1+list2
```

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Code

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```
20 10
```

[36]:
```
#**10. Write a program to merge two different lists.**
list1=[1,2,3]
list2=[3,4,5]

merged_list=list1+list2
print(merged_list)
list1.extend(list2)
print(list1)
merged_list1=[*list1,*list2]
print(merged_list1)

for i in list2:
    list1.append(i)
print(list1)
```

```
[1, 2, 3, 3, 4, 5]
[1, 2, 3, 3, 4, 5]
[1, 2, 3, 3, 4, 5, 3, 4, 5]
[1, 2, 3, 3, 4, 5, 3, 4, 5]
```

[37]:
```
list1=[1,2,3]
list2=[3,4,5]
list3=list(set(list1+list2))
print(list3)
```

```
[1, 2, 3, 4, 5]
```

[38]:
```
#**11. Write a program to read a random line in a file. (ex. 50, 65, 78th line)**

filename = "data.txt"
line_no = 50   # change to 65 or 78

with open(filename) as f:
    lines = f.readlines()
    print(lines[line_no - 1])
```


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[ ]:
```
#**12. Write a program to read random lines in a file. (ex. I want read all lines 10th to 15th line)**
filename = "data.txt"

start = 10
end = 15

with open(filename) as f:
    lines = f.readlines()
    for i in range(start-1, end):
        print(lines[i], end="")
```

[ ]:
```
#**13 Program to print the last "N" lines of a file.**
filename = "data.txt"
N = 5   # change this to any number you want

with open(filename) as f:
    lines = f.readlines()
    for line in lines[-N:]:
        print(line, end="")
```

[44]:
```
#**14. Write a program to check if the given string is Palindrome or not without using reversed method.**
s=12321
copy = s
rev =0
while s!=0:
    s1=s%10
    rev = rev*10 + s1
    s=s//10
if copy==rev:
    print(f"{copy} is palindrome")
else:
    print(f"{copy} is not palindrome")
```

```
12321 is palindrome
```

[45]:
```
h= "heleh"
copy = h
```

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Code JupyterLab Python

```
    print(f"{copy} is palindrome")
else:
    print(f"{copy} is not palindrome")
```

```
heleh is palindrome
```

[50]:

```
s="1231"
copy = s
rev =""

for ch in s:
    rev = ch + rev

if copy == rev:
    print(f'{copy} is palindrome')
else:
    print(f'{copy} is not palindrome')
```

```
1231 is not palindrome
```

[49]:

```
s = "abcba"
copy = s
rev = ""

for ch in s:
    rev = ch + rev

if copy == rev:
    print(f"{copy} is palindrome")
else:
    print(f"{copy} is not palindrome")
```

```
abcba is palindrome
```

```
s = "abcba"
if s == s[::-1]:
    print("Palindrome")
else:
```

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```
#**15 Write a program to search for a character in a given string and return the corresponding index.

s = "banana"
ch = "b"

indices = []
for i in range(len(s)):
    if s[i] == ch:
        indices.append(i)

print(indices)
```

[0]

```
# 16. Write a program to get the below output
# sentence = "hello world welcome to python programming hi there"
# d = {'h': ['hello', 'hi'], 'w': ['world', 'welcome'], 't': ['to', 'there'], 'p': ['python', 'programming'] }

sentence = "hello world welcome to python programming hi there"

words = sentence.split()
d = {}

for word in words:
    first = word[0]   # first letter

    # using if-else instead of 'not in'
    if first in d:
        d[first].append(word)
    else:
        d[first] = [word]   # create list with first word

print(d)


# 17 Write a program to replace all the characters with - if the character occurs more than once in a string**
#    python
```

```
# 17 Write a program to replace all the characters ...
# ```python
# my_string = 'hellohai' # O/P should be '-e--o-ai'

my_string = 'hellohai'
result = ""
seen = ""       # stores characters we have already seen

for ch in my_string:
    if ch in seen:
        result += "-"       # repeat → replace with -
    else:
        result += ch        # first time → keep character
        seen += ch          # mark as seen

print(result)
```

[ ]:

```
#**18 write a decorator that returns only positive values of subtraction**
def positive(func):
    def wrapper(a, b):
        return abs(func(a, b))    # abs() makes it positive
    return wrapper

@positive
def sub(a, b):
    return a - b

print(sub(10, 20))    # 10
print(sub(50, 30))    # 20
```

[ ]:

```
#**19 How to get the count of the number of instances of a class that is being created.**
class MyClass:
    count = 0    # class variable

    def __init__(self):
        MyClass.count += 1    # increase count every time object is created
```

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```
    return wrapper

@positive
def sub(a, b):
    return a - b

print(sub(10, 20))   # 10
print(sub(50, 30))   # 20
```

[53]:

```
#**19 How to get the count of the number of instances of a class that is being created.**
class MyClass:
    count = 0   # class variable

    def __init__(self):
        MyClass.count += 1   # increase count every time object is created

# creating objects
obj1 = MyClass()
obj2 = MyClass()
obj3 = MyClass()

print("Number of instances:", MyClass.count)
```

```
Number of instances: 3
```

[ ]:

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